Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/02/2005
PI Name:	Phillips, William D. Ph.D.		
Project Title:	Condensate Laboratory Aboard the Space Station (CLASS)		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Gaithersburg	**State:**	MD
Zip Code:	20899-8424	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	05/01/2000	**End Date:**	09/30/2006
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:	6	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: per telecon with asst, end date is 9/30/2006 (8/14/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Helmerson, Kristian (NIST) Lett, Paul (NIST) Porto, James (NIST)		
Grant/Contract No.:	NNC041A051		
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Task Description:	The ultralow temperatures and velocities achievable with Bose-Einstein condensation can be most fully exploited in a microgravity environment. Long interaction times, the manipulation of atoms with very small forces, and macroscopic quantum wave functions will be possible. To develop a Bose-Einstein condensate for use aboard the International Space Station. Experiments to be performed in the laboratory will develop the basic tools and experimental techniques needed to perform experiments in space. These may include matter wave interferometry with coherent matter wave, quantum reflection, and creation of picokelvin atom samples for future generation clocks. The combination of microgravity with Bose-Einstein condensation will allow the exploration of matter waves at extremely low temperatures and long (macroscopic) wavelengths. The research will enhance our understanding of BEC phenomena, and pave the road for the development of future technologies based on matter waves. In particular, it should be possible to create measurement instruments whose quality far surpasses that available today with earthbound techniques.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Matter wave interferometers are intrinsically more sensitive to certain influences, like rotation, than are optical interferometers, all other instrument parameters being equal. Rotation and acceleration sensing is essential to measurements for such applications as inertial navigation, gravitational mapping (which can be used in applications like navigation, prospecting, and searching for underground cavities. Ground based work on matter wave interferometers will impact both space and earth implementations. Spin off applications of condensates include quantum information, a subject with important national security implications, both for decryption and encryption. Applications of Bose-Einstein condensates to atomic clocks have potential earth based applications in navigation and communication.
Task Progress:	Work under this program covers a wide range of activity directed toward the development of tools for the coherent manipulation of Bose-Einstein condensates (BECs) for achieving the ultimate measurement benefits of BECs in the microgravity environment of the International Space Station (ISS) and other future missions. Among the major activities this year was moving our entire laboratory to NIST's new Advanced Measurement Laboratory, where state-of-the-art environmental conditions should allow improved measurements. All experiments achieved fully operational status by the end of the reported funding period. In our studies of deBroglie wave atom optics using BECs, we are now writing up a number of experiments on Bose-condensed atoms in moving and accelerating optical lattices. These experiments have demonstrated the coherent manipulation of atoms, and in particular the coherent acceleration by the addition of precise multiples of the photon momentum, a process with significant metrology implications. We have also started experiments in which multiple Bragg scattering or diffraction pulses applied to a BEC results in multiple-path interference between various components of the BEC that have been coherently separated and combined. These experiments are relevant to such areas as quantum random walks, kicked rotors, the classical-quantum cross-over, and quantum accelerator modes. In a quantum walk experiment we have observed a non-classical distribution after just a small number of steps, but technical difficulties with phase stability have prevented us from using a larger number of steps. In the kicked rotor experiment we have already observed some theoretically-predicted, but previously unobserved higher-order resonances in momentum transfer, and work is continuing. The sodium laser cooling apparatus that has traditionally studied collisions of ultracold atoms is currently being modified to allow for the creation of a Bose-Einstein condensate using an all-optical approach. In this approach, laser-cooled atoms collected in a magneto-optical trap (MOT) will be transferred into a far off-resonant laser dipole trap. The laser being used for the dipole trap is a 100 Watt fiber laser at 1.6 microns. Tight confinement, in order to achieve good thermalization of the trapped atoms by collisions, is facilitated by crossing the waists of two trapping beams. The evaporative cooling of atoms in the dipole trap is accomplished by lowering of the trap potential. The high power laser and imaging system are in place. We have succeeded in loading laser cooled atoms into the dipole trap and have seen preliminary evidence of evaporative cooling. An all optical approach to BEC will enable faster condensate production rates in an apparatus with excellent optical access. It will also eliminate the need for large magnetic trap coils used in most BEC set-ups that produce condensates with large numbers of atoms. Elimination of large magnetic trap coils will result in an apparatus that consumes less energy and therefore may be a more suitable approach aboard the ISS. We are producing rubidium condensates and loading them into optical lattices to investigate a number of interesting phenomena. Loading of the initial BEC into a two-dimensional optical lattice produces an array of one-dimensional (1-D) condensates. We previously demonstrated the presence and influence of spatial correlations in a 1-D Bose gas due to the interactions of the atoms. We are currently studying the transport of the 1-D degenerate Bose gas in the presence of a 1-D periodic optical potential and a weak harmonic confining potential. Our preliminary measurements indicate that the presence of even a weak periodic optical potential can result in substantial damping of the 1-D degenerate Bose gas oscillating in the weak harmonic confining potential. We are also studying the efficient creation of a three-dimensional (3-D) Mott-insulator state from the 1-D spatially correlated Bose gas. We are developing techniques to accurately evaluate the degree to which uniform filling of lattice sites is achieved in the 3-D Mott-insulator state.
Bibliography Type:	Description: (Last Updated: 06/01/2005)
Abstracts for Journals and Proceedings	Laburthe-Tolra, B.; O'Hara, K.M.; Huckans, J.; Rolston, S. L.; Phillips, W.D.; Porto, J. V. "Experimental Study of a Bose Gas in One Dimension" Laser Spectroscopy: Proceedings of the 16th International Conference on Laser Spectroscopy World Scientific eds. P. Hannaford, A. Sidorov, H. Bachor and K. Baldwin (2004) p. 124 , Oct-2004
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